What a Surrogate Cannot Tell You

Three omissions in a marginal-matched form/function discriminant,
and their consequences on the Riemann zeros and on lattice Yang–Mills

Amendments to FF06e

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Byline unset pending the author's decision — see **Contributions**, final page.

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Abstract

A *marginal-matched surrogate discriminant* asks, of each statistic in a battery, whether its value survives replacing the object by one drawn from the object's own one-point distribution with all higher structure destroyed. Survivors are said to read the distribution class (*form*); collapses to read the object (*function*). The instrument is a specialisation of surrogate-data testing [1] and was introduced to decide a question that arises whenever many statistics agree: whether their agreement is over-determination or redundancy. We observe that the operative definition omits three elementary conditions, state each as a proposition, and measure its consequences. **(1)** A witness invariant under the surrogate group has a *degenerate* null distribution — a point mass — so its standardised significance is 0/0; a small reported value is floating-point summation order presented as evidence. **(2)** A witness with bounded range has a *reachable ceiling* on significance; where that ceiling falls below the decision threshold, no object can be convicted and a verdict of *form* is unearned. **(3)** Non-invariance is necessary but not sufficient for a collapse, so *form* is a property of the pair (witness, object) and never of the witness. Recast, *form* is the statement that a witness *factors through the marginal*: a descent condition rather than a statistical one, which is why (1) is a corollary and not a finding. We implement a four-tier verdict and measure the amendments on two corpora sharing no subject matter: the first 2003 non-trivial zeros of ζ, where three of six verdicts change and the original form/function split otherwise reproduces; and a 6×6 $SU(2)$ lattice gauge configuration, where a battery of six witnesses is shown to carry at most three independent readings, two of them exactly invariant and two without power. We further localise a 0.4–1.0σ residual that FF06e attributes to re-interpolation: it requires piecewise-linear interpolation specifically and vanishes under exact inversion of the smooth counting function, so it is removable rather than a caveat to be carried. None of the three propositions is new as statistics; the contribution is that an instrument in use omits them, together with the measured size of each omission.

Status in this repository. This is a supporting document under `docs/`, outside the 35-paper canonical publication inventory and its build job, following the placement of the Yang–Mills comparison methodology paper. Its propositions and measurements are not part of the corpus author statement; see **Contributions** on the final page.

1 Claim boundaries

Status is separated from exposition throughout, and no claim below is stated more strongly in the text than in this table.

Claim	Status	Where
Prop. 1–3	elementary, proved	§5
A form witness is one factoring through the marginal	definitional recast	§4
Three of six verdicts change on the zeros	measured	§8
FF06e's form/function split otherwise reproduces	measured, smaller N	§8
Six lattice witnesses carry ≤ 3 readings	measured; 4 of 4 collapses structural	§9
The 0.4–1.0σ band requires linear interpolation	measured, one implementation	§10
Calibration is a precondition, not good practice	argued from a single failure mode	§7
Any claim about ζ itself	**none made**	§12

2 Introduction

A spectrum, a configuration, or a corpus admits no shortage of statistics. When many of them behave as expected, the usual reading is over-determination: independent witnesses, all agreeing, so the picture is robust. The opposite reading is available on the same evidence. If the statistics are not independent — if they are several lenses on one underlying response — their agreement is redundancy, and counting them as separate confirmations is a category error. Worse, if what they share is generic to the distribution class rather than specific to the object, their convergence reports only that the object belongs to that class, which is true of uncountably many objects with nothing else in common.

FF06e makes the question decidable with one operator. Replace the object by a *marginal-matched surrogate* — same one-point distribution, higher correlations destroyed — and ask of each witness whether its value survives. A survivor was reading the class (*form*); a collapse was reading the object (*function*), and the size of the collapse measures its influence. There is no parameter to tune, because the surrogate inherits the object's own distribution and the comparison is internal.

This is a specialisation of surrogate-data testing [1], whose structure is the same: specify a null, generate ensembles consistent with it, and compute a discriminating statistic on the original and on each surrogate. The specialisation is sharp and worth stating, because the null here is not a model fitted to the data but the data's own marginal, so the method is a permutation test with the object's empirical distribution as the null.

Our concern is the *definition* of the verdict, not the operator. We give three conditions it omits, each elementary, each with a measured consequence.

3 The discriminant

Definition 1 (Surrogate ensemble)**.** *Let X be the object, decomposed into components $x_1, \ldots, x_n$ carrying a distribution. Let $G = S_n$ act by permuting components. A* surrogate *is gX for g drawn uniformly from G. By construction every surrogate has the identical one-point distribution and no higher-order structure of X's.*

Definition 2 (Witness, significance)**.** *A* witness *is a map $W \colon X \mapsto \mathbb{R}$. With surrogate mean $\overline{W}$ and standard deviation σ_W over an ensemble of K draws, the* standardised significance *is*

$$z(W) \;=\; \frac{\left| W(X) - \overline{W} \right|}{\sigma_W}.$$

FF06e's Definition 2 declares W form *when $z(W)$ is small ($\lesssim 3$) and* function *when it is large.*

The instrument is strictly stronger than the control most evidence disciplines require, which is a null baseline measured on unrelated material. Unrelated material differs from the object in *both* its marginal and its structure, so a statistic that quiets down on it may be responding to either. Holding the marginal fixed leaves only structure.

4 Form as a descent condition

The statistical framing above obscures what is actually being asked. Let $\pi\colon X \mapsto X/G$ send an object to its orbit — equivalently, to its multiset of components, which is exactly its marginal.

Definition 3 (Factoring through the marginal)**.** *W factors through the marginal if there is $\widetilde{W}$ with $W = \widetilde{W} \circ \pi$, equivalently $W(gX) = W(X)$ for all $g \in G$.*

A form witness, in the limiting case, is precisely one that factors: its value is determined by the marginal and the fibre over that marginal carries no information it can see. A function witness does not factor; it sees the fibre. This is a local-to-global statement — the marginal is the quotient datum, and what the surrogate destroys is the fibre — and reading it that way makes the first amendment a one-line corollary rather than an empirical discovery.

5 Three omissions

Proposition 1 (Degenerate null)**.** *If W factors through the marginal, the null distribution of W is the point mass at $W(X)$. Hence $\overline{W} = W(X)$, $\sigma_W = 0$, and $z(W) = 0/0$ is undefined.*

Proof. $W(gX) = W(X)$ for every $g \in G$, so the pushforward of any measure on G is $\delta_{W(X)}$. □

The proposition is trivial; the consequence is not. FF06e's Definition 2 assigns *form* on the strength of a small z, and its headline table reports three witnesses at 0.4–1.0σ, while a construction remark states that the bare figure for those witnesses is "exactly zero by construction (symmetric-multiset invariants)." Both cannot be read at once, and the reported figure invites being read as a small real effect, which the remark denies.

A 0/0 evaluated in floating point returns whatever the summation order leaves behind. Measured on a lattice (§9), where two witnesses are provably invariant:

Witness	σ_W	computed z
mean $\mathrm{Re}\,U$ over links	6.5×10^{-16}	**1.4**
$\max\big\lvert\, \lVert U\rVert - 1 \,\big\rvert$	exactly 0	0.0

A z of 1.4 arising from summation order alone falls inside the band FF06e reports, on an object unrelated to the zeros and with no interpolation step present.

Proposition 2 (Reachable ceiling)**.** *Let W take values in $[a, b]$ and let the null distribution have mean μ and $\sigma_W > 0$. Then for every object X' with that null,*

$$z(W) \;\le\; C(W) \;:=\; \frac{\max(b - \mu,\ \mu - a)}{\sigma_W}.$$

If $C(W) < \tau$ for decision threshold τ, no object attains $z \ge \tau$.

Proof. $|W(X') - \mu| \le \max(b - \mu, \mu - a)$ since $W(X') \in [a, b]$. □

Where $C(W) < \tau$, a low z is a statement about the instrument's resolution and carries no information about the object; declaring *form* attributes to the object what belongs to the test. Measured: a Wilson loop trace on a 6×6 lattice, with range $[-1, 1]$, has σ_W between 0.11 and 0.45, giving C below 5 and in places below 2 against $\tau = 3$. A synthetic case isolates the mechanism: a witness returning one component of a sequence spanning $[-1, 1]$ has σ_W equal to the component spread (≈ 0.58) and $C \approx 1.7$, so it can never be convicted however structured the object.

FF06e's amendment of 24 September 2026 asks for exactly this condition — "a null result must retain power and resolution limits" — but Definition 2 does not encode it, so a reader following the definition cannot apply the amendment.

Proposition 3 (Non-invariance is not sufficient)**.** *Non-invariance of W is necessary for $\sigma_W > 0$ but does not imply $W(X)$ is separated from the null. There exist non-invariant W and objects X with $z(W)$ arbitrarily small.*

Proof. By example. Let W count ascents, $W(x) = \#\{i : x_{i+1} > x_i\}$, which is not G-invariant. For X a discretised sine period of length n, $W(X)$ is $\lfloor n/2 \rfloor$ or thereabouts, since a sine ascends on exactly half its period; the null mean is $(n-1)/2$. Measured at $n = 60$: $W(X) \in \{29, 30\}$, null mean 29.5, $z = 0.34$. The same W on a monotone sequence of the same length gives $W = n - 1$ and large z. □

Hence *form* is a property of the pair (witness, object). The practical consequence is a reading of FF06e's own results: its distributional witnesses are form *on the first* 10^5 *zeros*, which is what was measured, and not form as a standing property — they are serviceable statistics on an object whose spacing distribution differs from its own shuffle.

Remark 1. None of the three is novel as statistics. Permutation tests require a statistic that varies under the null, and any test requires power; both are standard. The contribution is that an instrument in use omits them, and the measured size of each omission.

6 A four-tier verdict

The propositions give a decision order, applied per witness, with ε a floating-point floor scaled to $|W(X)|$:

1. $\max_i |W_i - W(X)| \le \varepsilon$ over the ensemble ⇒ FORM_EXACT; report no z (Prop. 1).
2. $\sigma_W \le \varepsilon$ but $W(X)$ differs ⇒ FUNCTION, z unbounded. Every surrogate agrees with every other and not with the object.
3. W bounded and $C(W) < \tau$ ⇒ NO_POWER (Prop. 2). Unresolved, not form.
4. Otherwise FUNCTION if $z \ge \tau$, else FORM — a verdict about this pair (Prop. 3).

A separate check detects a different redundancy: witnesses pairwise affine across three or more objects are one witness in two costumes, and counting both double-counts.

7 Calibration is a precondition

FF06e calibrates its engine against known invariants and known refractions with negative controls and decoys, "so that the engine could not become its own refraction." We find this cannot be relaxed to good practice, because one failure mode has no other detector.

The caller must supply the surrogate construction, since only the caller knows what the object's marginal is; this is the sole irreducibly domain-specific decision in the method. If the supplied construction does not preserve the marginal, it is randomisation, and then every witness reads FUNCTION for a reason having nothing to do with the object. Measured: replacing the permutation with fresh random values convicts the sample variance — a provable invariant — as FUNCTION. No internal consistency check catches this and nothing about the output looks wrong. The only detector is the requirement that a known invariant return FORM_EXACT.

The protocol, as implemented:

1. Known invariants ⇒ FORM_EXACT, z withheld.
2. Known non-invariant witnesses on a structured object ⇒ FUNCTION.
3. A decoy returning noise independent of the object ⇒ not FUNCTION.
4. A deliberately marginal-breaking surrogate ⇒ must convict an invariant.
5. Fewer than two surrogates ⇒ refuse to run.

8 Corpus I: the zeros of ζ

Instrument. Riemann–Siegel Z with two remainder terms [3], zeros by sign change and bisection, 2003 zeros on $t \in (20, 2520)$, unfolded by the Riemann–von Mangoldt term. Anchored before use: θ at the first Gram point (17.8455995408 against 17.8455995405); the first ten zeros against published values, worst deviation 2.5×10^{-3}; and the counts $N(50) = 10$, $N(100) = 29$, $N(300) = 138$, $N(1000) = 649$, all exact, with $S(T) = N(T) - \langle N(T) \rangle$ bounded at $+0.58, -0.00, +0.29, +0.38$. Positions carry 2.5×10^{-3} — the remainder is asymptotic and $t \approx 14$ is where it is weakest — while the counts are exact. The distinction matters: a missed zero is invisible locally and would appear only as $S(T)$ drifting.

Result. Surrogate by permuting unfolded spacings; $K = 40$; the arithmetic witness is the prime-resonance power $\sum_{p \le 29} \langle \cos(\gamma \log p) \rangle^2$ on actual heights, as in FF06e.

Witness	σ_W	z	$C(W)$	Tier
arithmetic, $p \le 29$	3.51×10^{-4}	**499.8**	—	FUNCTION
lag-1 spacing correlation	2.16×10^{-2}	**17.6**	46.6	FUNCTION
spacing mean deviation	6.97×10^{-16}	withheld	—	FORM_EXACT
spacing variance	1.67×10^{-16}	withheld	—	FORM_EXACT
Wigner-shape L^2	exactly 0	withheld	—	FORM_EXACT

FF06e's split reproduces: the arithmetic witness carries the overwhelming majority of the object-specific influence and the distributional witnesses carry none. Prop. 1 changes the last three verdicts from *form at* 0.4–1.0σ to *exactly invariant, significance withheld.* An amendment altering no verdict would be a stylistic preference.

A check on amplification. FF06e's decisive argument is that the function signal sharpens with N rather than averaging down, reporting $z_{\text{arith}} = 3030$ at $N = 2 \times 10^4$ and 11 497 at $N = 10^5$. The exact two-point exponent is 0.829, which extrapolated a decade below the fitted range predicts 450; we measure 499.8, i.e. 11% high. (An earlier version of this paragraph said 4%, using the exponent rounded to 0.8, which moves the prediction to 481. The rounding flattered the comparison. A

two-point fit carries no uncertainty estimate, so 11% a decade outside the fitted range is agreement rather than a discrepancy.) Its paired caveat holds in the stated direction: the raw value is *larger* at smaller N (0.1762 here against 0.0922 and 0.0636), so significance and magnitude move oppositely. A third point on a curve fitted from two is the strongest evidence here that the amplification is structure being resolved.

9 Corpus II: lattice $SU(2)$

A second corpus is needed because the amendments are claims about a method, and a method validated on one object is accommodated to it.

Object. A 6×6 periodic lattice carrying a unit quaternion $U \in SU(2)$ on each link — a Wilson lattice gauge configuration [4]. The marginal is the multiset of link rotors; the surrogate permutes which site each link occupies. Spatial adjacency is what gauge structure consists of, a plaquette being the ordered product around a face, so adjacency is the right structure to destroy.

Witness	smooth configuration	relaxed two-phase	note
mean $\mathrm{Re}\,U$ (links)	FORM_EXACT	FORM_EXACT	invariant; also gauge-dependent
$\max \lvert\lVert U\rVert - 1\rvert$	FORM_EXACT	FORM_EXACT	invariant
mean $\mathrm{Re}\,P$ (faces)	$z = 6.6$ FUNCTION	$z = 2.0$	
Wilson action at $\beta = 1$	$z = 6.6$	$z = 2.0$	affine to the above
loop trace	$z = 1.1$	$z = 0.8$	NO_POWER
order sensitivity	$z = 0.6$	$z = 0.2$	NO_POWER

Six witnesses carry at most three independent readings. Two are exactly invariant (Prop. 1). The Wilson action at $\beta = 1$ equals $wh\,(1 - \langle \mathrm{Re}\,P \rangle)$, hence is affine to the face mean, and their z agree to nine decimals because they must. Two have $C(W) < \tau$ (Prop. 2). All four collapses are structural rather than statistical.

Remark 2. A corollary worth recording. The link mean has no consumer anywhere in the implementing codebase, and the instrument supplies the reason: it is gauge-*dependent* — a gauge transformation moves it while the face mean holds to nine decimals — and surrogate-invariant. Gauge-dependent together with invariant exhausts the ways a lattice witness can fail to inform.

That the same three propositions bite on two corpora sharing no subject matter is the evidence that they are amendments to the method rather than to one dataset.

10 A removable artifact in the surrogate construction

FF06e attributes its 0.4–1.0σ form-witness band to its surrogate path: permute unfolded spacings, re-accumulate, then interpolate back onto the reference grid. The attribution is correct as to mechanism and localises further than stated. Two routes from unfolded positions back to heights:

Route	spacing mean dev.	spacing variance	Wigner L^2
Analytic: bisect the monotone $\langle N \rangle$	6.2×10^{-16}	4.4×10^{-15}	exactly 0
Piecewise-linear between samples	1.9×10^{-16}	$\mathbf{1.03 \times 10^{-5}}$	$\mathbf{3.51 \times 10^{-4}}$

The artifact requires interpolation between knots, not the round trip. Inverting the smooth counting function by bisection restores exact invariance. A caveat a better pipeline removes should be removed rather than documented.

We do not compare the resulting z against FF06e's 0.4–1.0: ours are 0.16 and 0.01, and z here is a ratio of one small shift to another, dominated by grid, N and interpolation detail. The finding is σ_W moving from machine epsilon to 10^{-5}; its standardisation is not a quantity worth comparing across implementations.

11 Discussion

The amendments cost something, and the cost is worth naming. Prop. 2 converts some former *form* verdicts into *unresolved*, which is a loss of apparent information. The trade is favourable for an asymmetric reason: an unresolved verdict can be improved by a better witness or a larger object, whereas a false *form* verdict cannot be improved, because it does not know it is wrong.

Prop. 1 replaces a number with a refusal. This is the amendment most likely to be resisted, since a table with an omission looks weaker than a table of figures. It is the other way around: the figure was never a measurement, and a reader who takes 1.0σ for a small real effect has been misled by the presentation rather than by the instrument.

Read as descent (§4), the three propositions say something compact. The surrogate construction is a choice of quotient. A witness either factors through that quotient or it does not; if it does, there is nothing to test, and if it does not, whether it *detects* the fibre is a separate question of power and of the particular object. Conflating "does not factor" with "detects" is the single error behind all three omissions.

12 Scope and claim boundaries

What a function verdict establishes. That the witness responds to structure beyond the object's one-point distribution, relative to that object and that surrogate construction. It does *not* establish that the witness is correct, meaningful, or evidence for any proposition about the object. FF06e is explicit on this and it bears repeating, because a large z reads as a result: a witness may respond strongly to the object and be the wrong statistic.

No claim is made about ζ. Nothing here concerns the Riemann hypothesis, the construction of a self-adjoint operator, or off-line behaviour. Corpus I is used as an object with known structure on which to measure a method.

No claim is made about $SU(2)$ gauge theory. Corpus II is a finite lattice configuration used for the same purpose.

Where the method stops. It requires a surrogate preserving the marginal while destroying correlation. Where no such construction exists — where the object cannot be quotiented to separate its response from the instrument's — the form/function question is well posed and not answerable by this operator. That is the tool's edge, not a defect in results obtained inside it.

Single-implementation results. §10 rests on one implementation of each route. The qualitative finding (machine epsilon against 10^{-5}) is robust to implementation; the particular figures are not.

13 Reproduction

Seed 20260423 throughout. The implementation is the `AISO-UI-for-Ai` tree (`frontend/src/shuffle-knife.ts`, `frontend/src/riemann-siegel.ts`, `frontend/src/rotor-field.ts`, `frontend/src/rotor-dynamics.ts`). The tree is public at https://github.com/tensorrent/AISO-UI-for-Ai; the numbers and figures in this document were produced on branch `claude/excel-memory-offloading-9ezwrv`, and the four instrument files, the scripts and the lattice test are byte-identical from the generating commits (`086a0e3`, `64d9658`) through commit `3a8e610`, which is the pin. Clone the repository and check it out at that commit,

```
git checkout 3a8e610a80575da13ee68649c7a92ba3690be140
```

then `npm install` in `frontend/` and run the commands below from that directory.

Command	Produces
`npx vitest run tests/shuffle-knife.test.ts`	the calibration protocol, §7
`npx vitest run tests/riemann-siegel.test.ts`	the instrument anchors, §8
`npx tsx scripts/riemann-witnesses.mts`	Corpus I, both surrogate routes
`npx vitest run tests/rotor-dynamics.test.ts`	Corpus II

Scripts depending on an anchored instrument exit non-zero if an anchor fails, so no table in this paper can arise from a run whose instrument was not validated.

A Dead ends and self-corrections

Recorded because the method's own discipline requires it, and because two of the results above exist only as the residue of an error.

1. **The ascent count was predicted to read function.** Asserted in calibration; the assertion failed and the verdict was correct. This became Prop. 3, the amendment with the widest reach.
2. **An analytic round trip was expected to reproduce FF06e's band.** It does not — σ_W stays at machine epsilon. The negative result showed "re-interpolation" to be too coarse a diagnosis and produced §10.
3. **Two implementations of a Condorcet cyclicity figure disagreed by** 3.4σ (49.2% against 46.5% at five candidates, 4000 trials each) — a spread as large as a discrepancy then being claimed against a third party's published number. The claim was withdrawn until two seeds at 20 000 trials and three classical anchors settled it. The disagreement is retained as the apparatus band it was.
4. **A mean was read as a partial collapse.** On the two-phase lattice configuration the face mean is 0.263775, which is the exact count-weighted average of 28 faces at 0.05342509 and 8 at exactly 1, and is attained by no face. The surrogate test then shows the mean sits $\approx 2\sigma$ from its own null on that object: the domain structure is invisible to it and was visible only in the histogram. A scalar that cannot see what its own relation shows.

References

[1] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, J. D. Farmer, *Testing for nonlinearity in time series: the method of surrogate data*, Physica D **58** (1–4), 77–94, 1992. doi:10.1016/0167-2789(92)90102-S.

[2] B. Wallace, *Spectral Rigidity and Shuffled Spacing Discriminant: Categorising Spectral Witnesses by What They Respond To* (FF06e), TensorRent, 2026; amendment of 24 September 2026.

[3] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974 (Riemann–Siegel formula, Ch. 7).

[4] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D **10**, 2445–2459, 1974.

Contributions. The discriminant, the form/function vocabulary, the reactional versus response-driven framing, and the finding that a battery of agreeing witnesses may be one reading refracted are due to FF06e [2]. The three propositions of §5, the descent recasting of §4, the four-tier verdict, both corpora, and the interpolation finding of §10 are this record's. All computation was performed by the implementation cited in §13 and is reproducible from it; no figure in this paper is quoted from memory or from a source not opened.